



Wireless Networks

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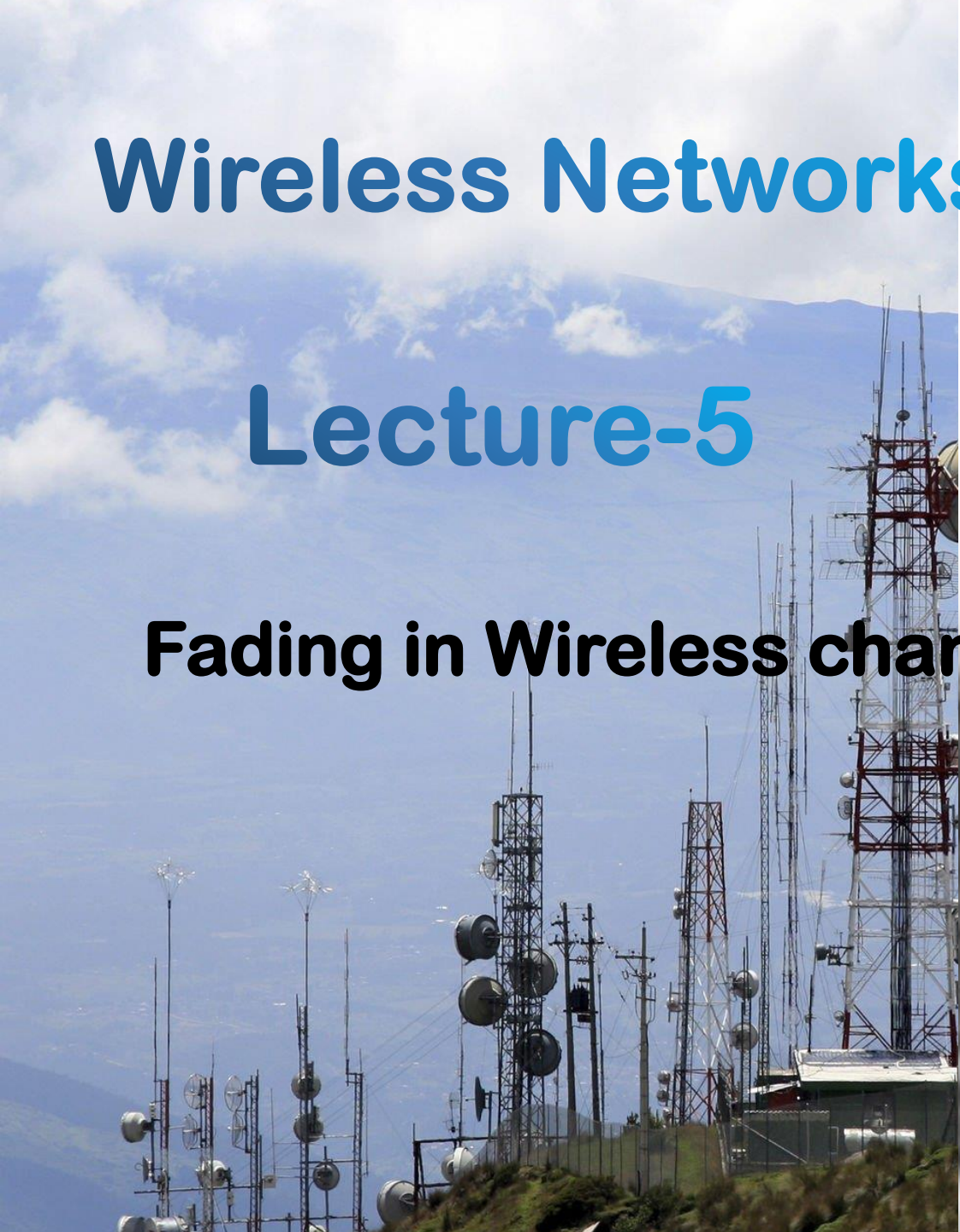
Suez Canal University

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Wireless Networks

Lecture-5

Fading in Wireless channel



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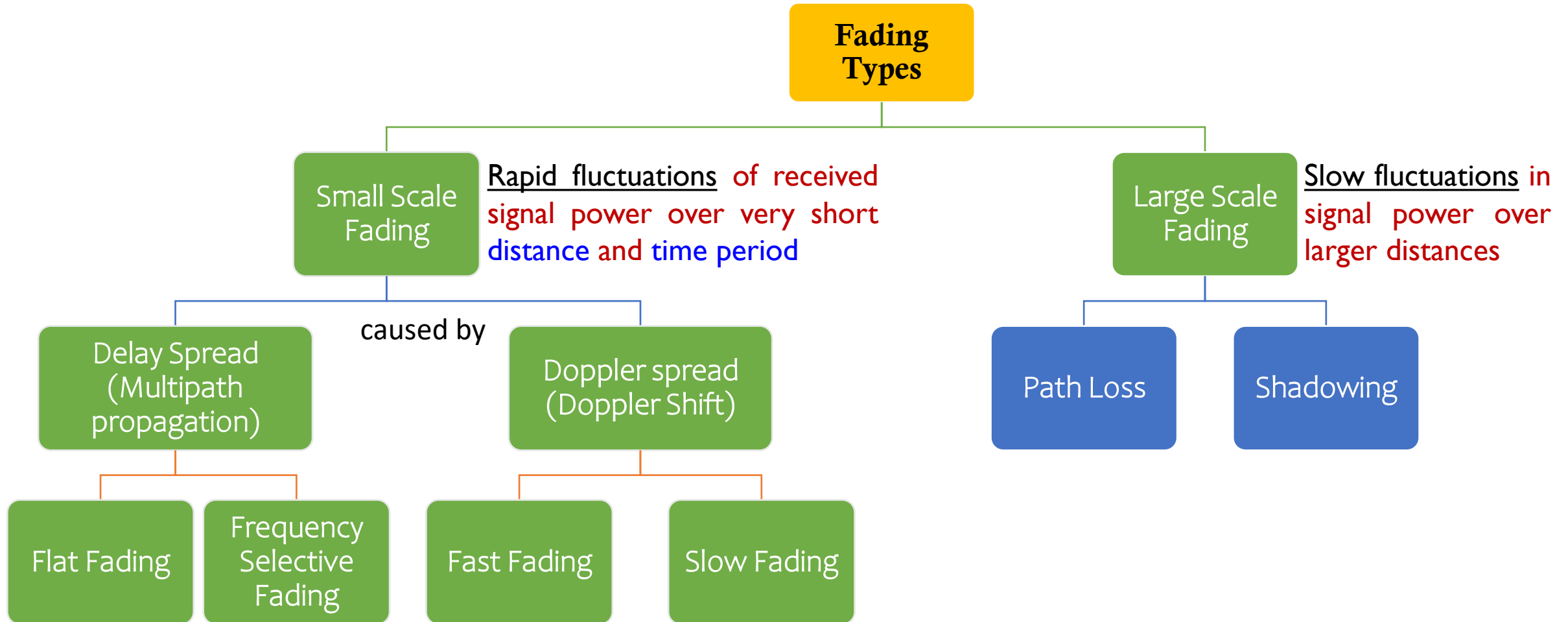


Fading in wireless transmission: Definition

Fading is the **fluctuation** in received signal power caused by **variations** in the transmission channel, resulting in signal **degradation** and potential data transmission **errors**. The channel **variations** arise from factors such as:

- 1) Multipath propagation.
- 2) Doppler shift.
- 3) Environmental obstructions like buildings, trees, and moving objects.

Fading Classification

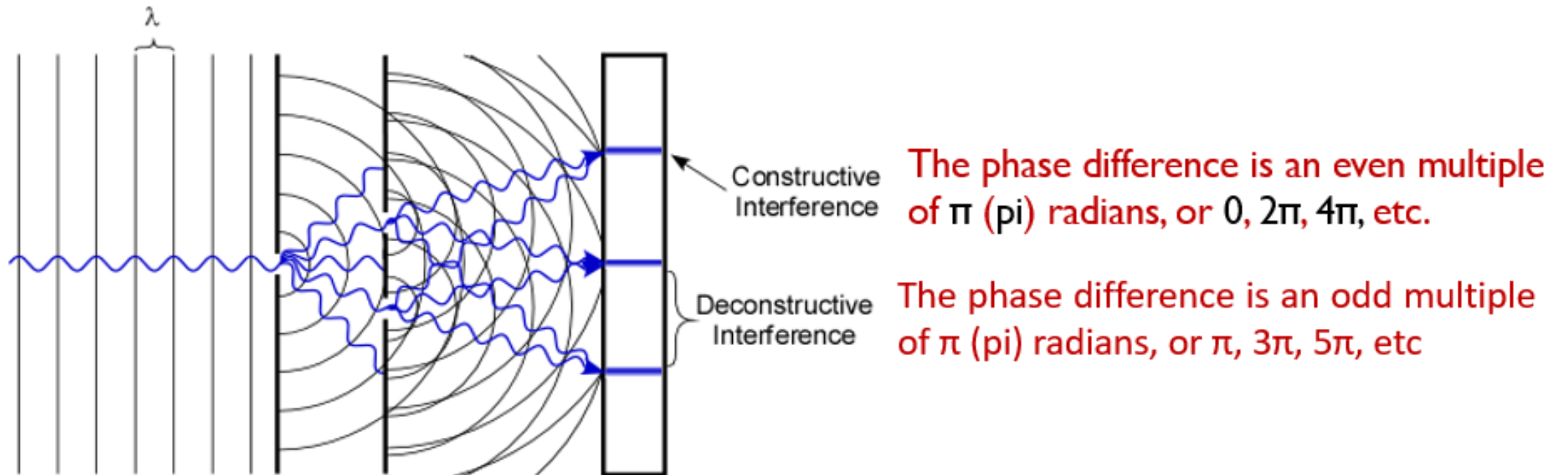


Small-Scale Fading: Causes

I) Multipath propagation

In multipath propagation, **multiple copies** of the **same signal** travel along **different paths** to the receiver, arriving with distinct **delays**, **phases**, and **amplitudes**, leading to fading effects.

a) Constructive and Destructive Interference – Constructive and destructive interference occur when multiple signals overlap, with the outcome depending on their relative phases. This typically leads to rapid fluctuations in signal strength.

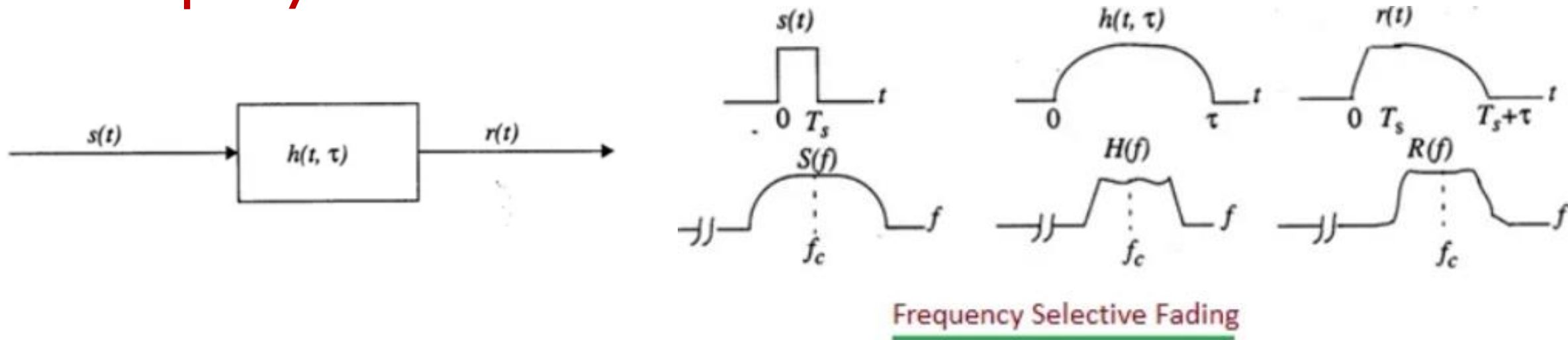


Small-Scale Fading: Causes

I) Multipath propagation

b) Frequency Selective

Different frequency components of a transmitted signal experience different levels of attenuation due to multipath propagation. This happens when the **signal bandwidth** B is greater than the **coherence bandwidth** B_c of the channel (i.e. $B > B_c$) leading to **frequency distortion**.



Small-Scale Fading: Causes

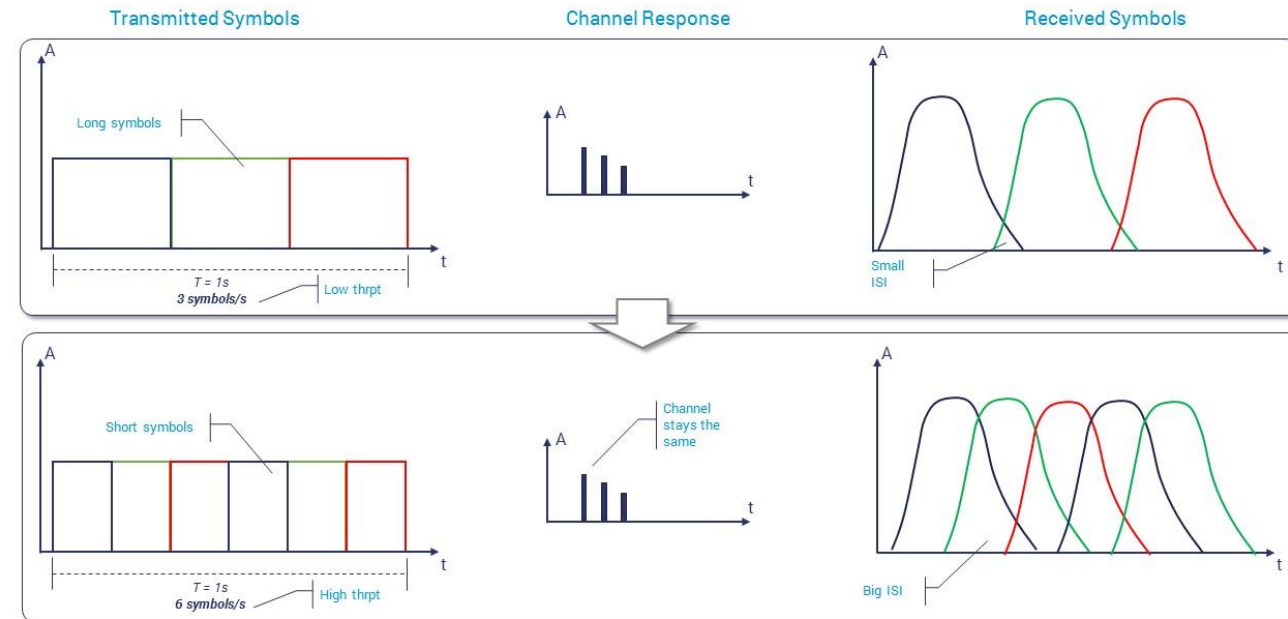
I) Multipath propagation

c) **Time Dispersion (Delay Spread)** – Time dispersion occurs when multiple copies of a transmitted signal arrive at the receiver at different times causing **time dispersion** (تشتت الوقت). The difference between the arrival times of the earliest and latest received signals is called **delay spread**.

❖ Effects of Time Dispersion

- ☒ **Inter-Symbol Interference (ISI)** – previous symbols interfere with current ones, leading to signal distortion.
- ☒ **Frequency Selective Fading** .
- ☒ **Signal Fading** –constructive or destructive interference, causing variations in signal strength.

Inter-symbol Interference



Small-Scale Fading: Causes

I) Multipath propagation

○ Time Dispersion Parameters

I) **Excess Delay (τ)** – Measures how much later a multipath component arrives compared to the direct path.

Mathematically:

$$\tau = t_{\text{multipath}} - t_{\text{direct}}$$

where:

- $t_{\text{multipath}}$ is the arrival time of a reflected or scattered signal.
- t_{direct} is the arrival time of the direct signal.

Small-Scale Fading: Causes

I) Multipath propagation

○ Time Dispersion Parameters

2) Mean Excess Delay ($\bar{\tau}$)

The **Mean excess delay** ($\bar{\tau}$) is the mean delay of multipath components, weighted by their respective power levels. It represents the **center of mass** of the power delay profile (PDP).

$$\bar{\tau} = \frac{\sum_i P_i \tau_i}{\sum_i P_i}$$

where:

- P_i = Power of the i -th multipath component
- τ_i = Excess delay of the i -th multipath component
- $\sum_i P_i$ = Total received power

Small-Scale Fading: Causes

I) Multipath propagation

○ Time Dispersion Parameters

2) Mean Excess Delay ($\bar{\tau}$)

Scenario: A mobile user is communicating with a base station in an **urban environment**. The transmitted signal undergoes **multipath propagation** due to reflections from nearby buildings. At the receiver, three multipath components are detected with the following **delays and power levels**:

Path	Delay (τ_i) in μs	Power (P_i) in mW
1	0.5 μs	4 mW
2	2.0 μs	3 mW
3	4.5 μs	1 mW

Solution

$$\bar{\tau} = \frac{(4 \times 0.5) + (3 \times 2.0) + (1 \times 4.5)}{4 + 3 + 1} = \frac{2 + 6 + 4.5}{8} = \frac{12.5}{8} = 1.56 \mu\text{s}$$

Interpretation: Urban areas typically have higher delay spreads due to multiple reflections, compared to open areas.

Small-Scale Fading: Causes

I) Multipath propagation

○ Time Dispersion Parameters

3) Root Mean Square (RMS): It measures how much the received signal is spread out in time. A higher RMS delay spread leads to inter-symbol interference (ISI), affecting system performance.

$$\sigma_{\tau} = \tau_{\text{rms}} = \sqrt{\frac{\sum P_i (\tau_i - \bar{\tau})^2}{\sum P_i}}$$

Scenario:

A **5G mobile user** is communicating with a base station in a **dense urban area** where multipath propagation occurs due to reflections from buildings and vehicles. The receiver detects the following multipath components:

Path	Delay (τ_i) in μs	Power (P_i) in mW
1	0.4 μs	5 mW
2	1.5 μs	3 mW
3	3.2 μs	2 mW

Measures how much the received signal is spread out in time.

Small-Scale Fading: Causes

I) Multipath propagation

○ Time Dispersion Parameters

Step 1: Compute the Average Excess Delay ($\bar{\tau}$)

$$\begin{aligned}\bar{\tau} &= \frac{\sum P_i \tau_i}{\sum P_i} = \frac{(5 \times 0.4) + (3 \times 1.5) + (2 \times 3.2)}{5 + 3 + 2} \\ &= \frac{2 + 4.5 + 6.4}{10} = \frac{12.9}{10} = 1.29 \mu\text{s}\end{aligned}$$

Step 2: Compute the RMS Delay Spread (τ_{rms})

Formula:

$$\tau_{\text{rms}} = \sqrt{\frac{\sum P_i (\tau_i - \bar{\tau})^2}{\sum P_i}}$$

Computation in One Line:

$$\begin{aligned}\tau_{\text{rms}} &= \sqrt{\frac{(5 \times (0.4 - 1.29)^2) + (3 \times (1.5 - 1.29)^2) + (2 \times (3.2 - 1.29)^2)}{5 + 3 + 2}} \\ &= 1.07 \mu\text{s}\end{aligned}$$

Interpretation:

→ A lower **RMS delay spread** indicates

less time dispersion, reducing **ISI**.

→ High **RMS delay spread** means

multipath signals arrive with **large**

delays, which can cause **fading and**

signal distortion.

→ Wireless systems like **5G** use

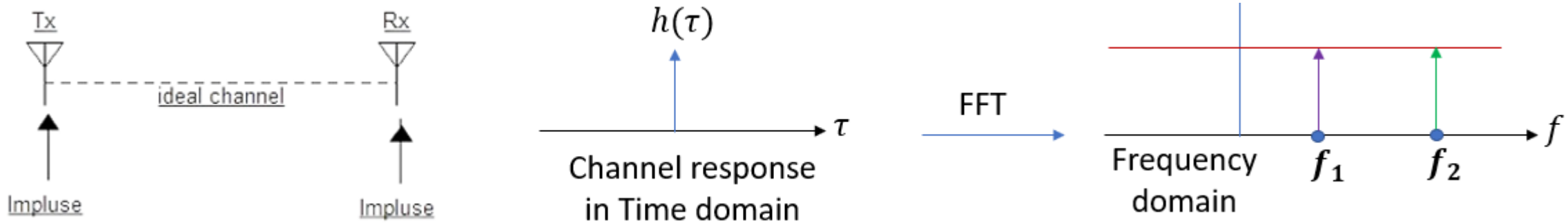
equalization techniques to

counteract this effect.

Small-Scale Fading

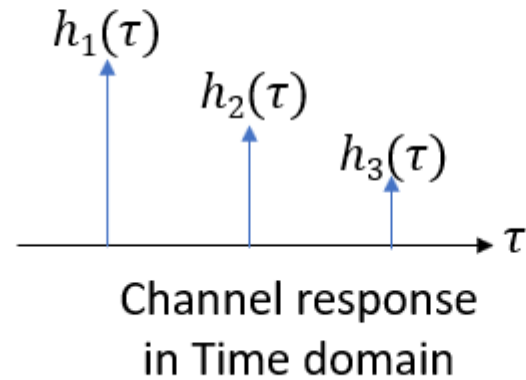
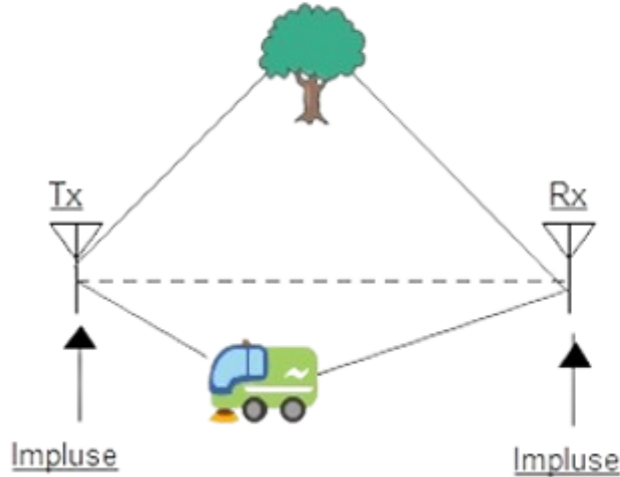
☑ Coherence Bandwidth, B_c

- If there is only one LoS path, the channel response will be represented as one impulse.
- So, if you send impulse, you will get impulse at receiver.
- If you send two signals at two different frequencies f_1 and f_2 , the channel response will be the same for them.
- This means that, whatever the transmitted signals over any frequency, the channel response will manipulate them with same processing.

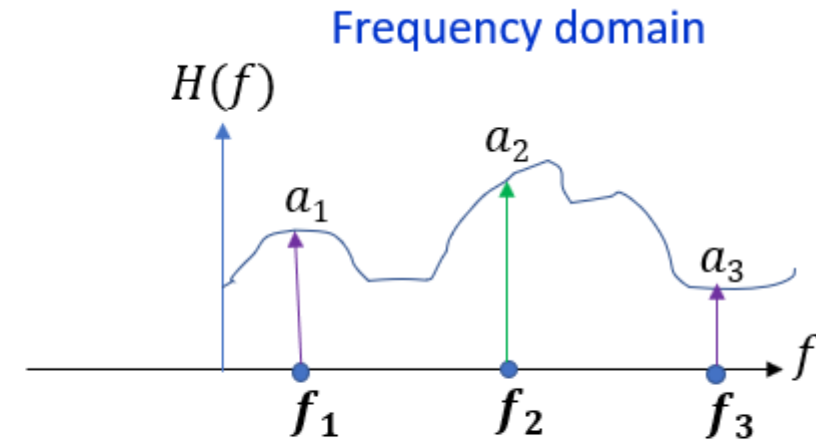


Small-Scale Fading

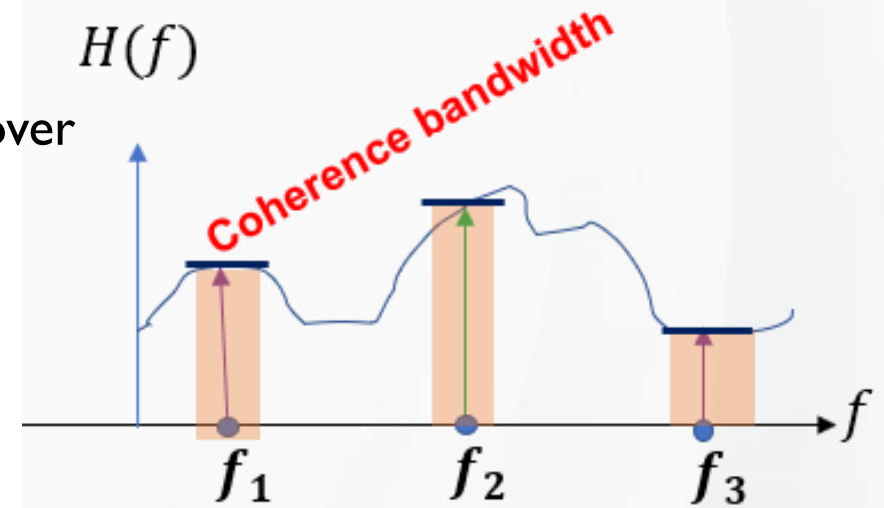
☑ Coherence Bandwidth, B_c



FFT



- Three paths **result in** three distinct impulse response channels. In the time domain, each frequency is processed differently with a unique coefficient a_i .
- **Coherence bandwidth** refers to the range of frequencies over which the channel response **remains nearly constant**.



Small-Scale Fading

☑ Coherence Bandwidth, B_c

Common Approximations for Coherence Bandwidth (B_c)

Two commonly used formulas:

1. Approximate Estimate:

$$B_c \approx \frac{1}{5\tau_{\text{rms}}}$$

- Gives a **higher** value of B_c , meaning the channel is assumed to be **less dispersive**.
- Suitable for **ideal or moderate** fading conditions.

2. Conservative Estimate:

$$B_c \approx \frac{1}{50\tau_{\text{rms}}}$$

- Gives a **lower** value of B_c , meaning the channel is assumed to be **highly dispersive**.
- Used in **real-world scenarios** where the channel may be more unpredictable.
- Ensures **robust communication design**, minimizing **frequency-selective fading** issues.

Small-Scale Fading

☑ Coherence Bandwidth, B_c

Scenario:

A **4G LTE base station** is deployed in a **suburban environment**, where a mobile user experiences **multipath propagation** due to reflections from trees, houses, and vehicles. The channel characteristics are measured, and the **RMS delay spread** is found to be $\tau_{rms} = 2.5 \mu s$. Calculate the coherence bandwidth.

Solution

1. Approximate Estimate:

$$B_c = \frac{1}{5 \times 2.5 \times 10^{-6}} = \frac{1}{12.5 \times 10^{-6}} = 80 \text{ kHz}$$

2. Conservative Estimate:

$$B_c = \frac{1}{50 \times 2.5 \times 10^{-6}} = \frac{1}{125 \times 10^{-6}} = 8 \text{ kHz}$$

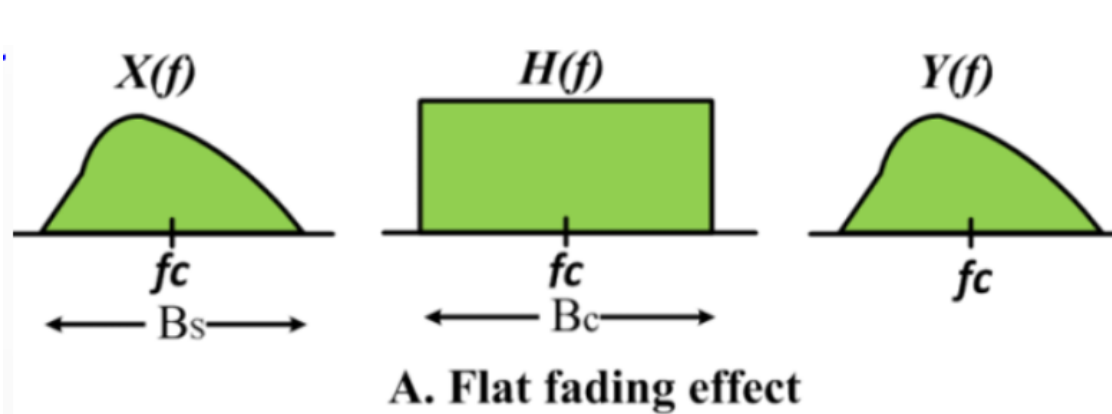
In practical wireless design, the conservative estimate (8 kHz) is often preferred to ensure a reliable system that can handle fading effects better.

Interpretation:

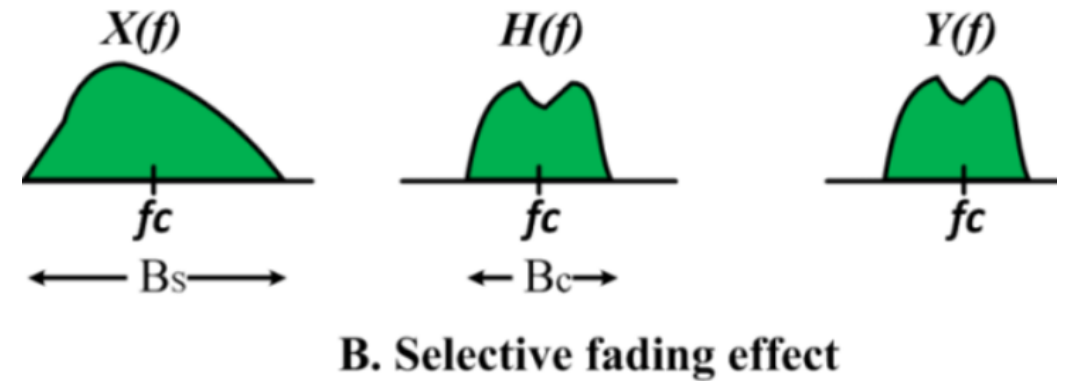
→ In this scenario, the conservative estimate ($B_c = 8 \text{ kHz}$) suggests that signals with **bandwidths larger than 8 kHz** may experience **frequency-selective fading**.

Small-Scale Fading Types

The types of small-scale fading caused by multipath propagation



if the signal $B_s < B_c \rightarrow$ **Flat Fading**
(all frequency components experience similar fading).



If the signal bandwidth $B_s > B_c \rightarrow$ **Frequency-Selective Fading**
(different frequency components experience different fading).

Small-Scale Fading Types

❖ Relation Between Symbol Time T_s and Standard Deviation of Delay Spread σ_τ

- ☑ In wireless communication, **Inter-Symbol Interference (ISI)** occurs when delayed versions of a transmitted signal overlap with subsequent symbols.
- ☑ The relationship between **symbol time T_s** and **the standard deviation of delay spread σ_τ** determines whether a system experiences **flat fading** or **frequency-selective fading**

→ Flat Fading Condition:

$$T_s \gg 10 \sigma_\tau$$

→ Frequency-Selective Fading Condition:

$$T_s \ll 10 \sigma_\tau$$

Small-Scale Fading

- Coherence time, T_c

It represents the time interval during which a wireless channel does not experience significant changes due to fading or movement of the transmitter, receiver, or surrounding environment.

$$T_c \approx \frac{1}{f_d}$$

$$f_d = \frac{vf_c}{c}$$

where f_d is the Doppler spread, which depends on the relative velocity (v) between the transmitter and receiver, the carrier frequency (f_c), and the speed of light (c).

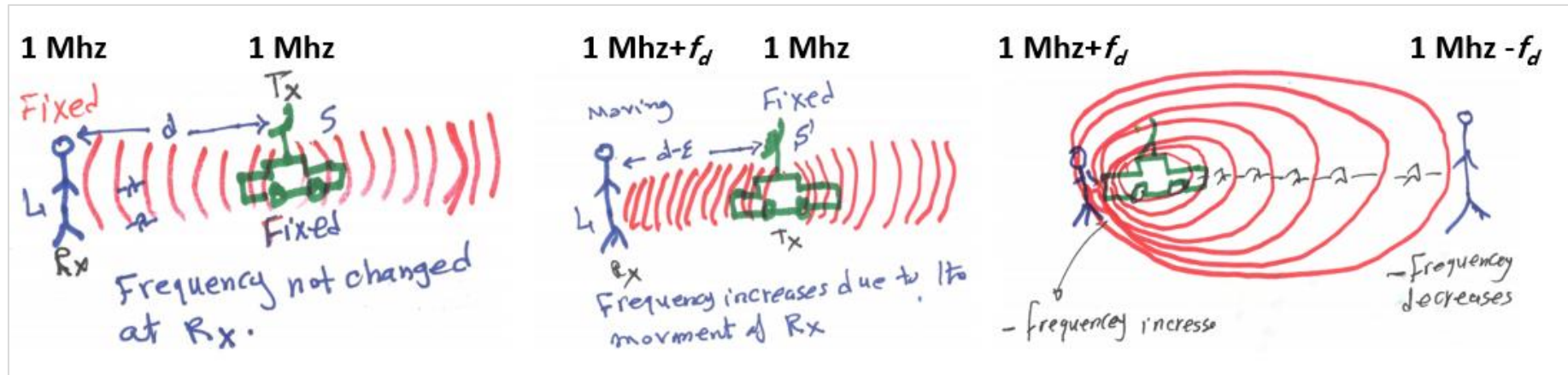
Small-Scale Fading: Causes

2) Doppler Spread

✓ **Doppler shift:** Describe how the frequency of a received signal changes if the transmitter, receiver, or both are moving.

→ If the source moves toward the receiver, the frequency increases (positive shift).

→ If the source moves away from the receiver, the frequency decreases (negative shift).



✓ **No Doppler effects if :**

- 1) Tx and Rx are fixed
- 2) Tx and Rx are Fixed while medium moving
- 3) Tx and Rx are moving with the same speed
- 4) Tx and Rx are moving in mutually perpendicular direction

Small-Scale Fading: Causes

2) Doppler Spread

Doppler Spread: When multiple scatterers (e.g., buildings, trees) reflect the transmitted signal, each path can experience a different Doppler shift. This results in a **spread** of frequencies rather than a single shift.

- a) If the Doppler spread is high, it leads to **fast fading**, where the signal varies quickly over time.
- b) The **coherence time** T_c (how long the channel stays constant) is approximately given by

$$T_c \approx \frac{1}{f_{max}}$$

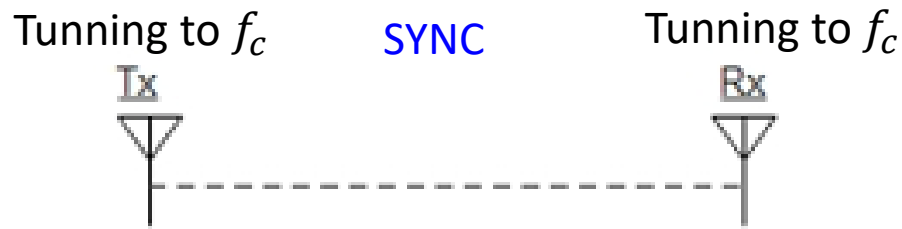
where f_{max} is the maximum Doppler shift.

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift effects

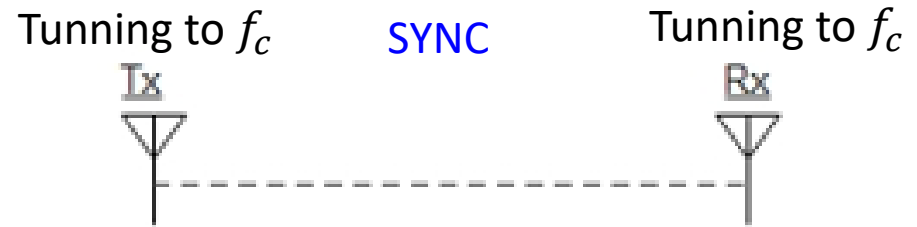
- Doppler shift **alters the carrier frequency**, causing **mismatches** between the transmitter and receiver frequencies.



No doppler shift

Modulation $a(t) e^{j2\pi f_c t}$

Demodulation $a(t) = a(t) e^{j2\pi f_c t} \times e^{-j2\pi f_c t}$



Doppler shift

Modulation $a(t) e^{j2\pi f_c t}$

Demodulation $a(t) = a(t) e^{j2\pi(f_c+f_d)t} \times e^{-j2\pi f_c t}$
 $= a(t) e^{j2\pi f_d t}$

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift calculation

- λ : Carrier wave length
- f_c : Carrier Frequency
- θ_i : Angle between ME movement direction and the direction of the incoming waves from antenna.
- After Δt , the ME moves distance $v\Delta t$
- After Δt , the change in the propagation path is given as :

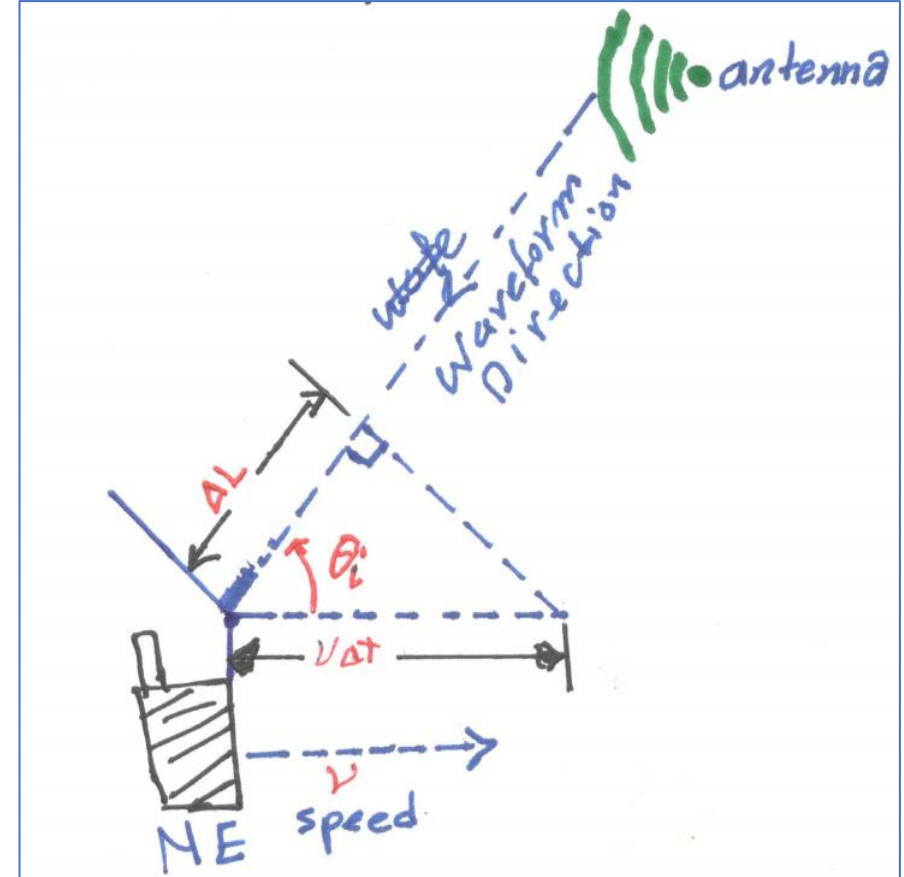
$$\Delta L = v\Delta t \cos(\theta_i)$$

- The phase change $\Delta\phi$ due to ΔL is given as

$$\Delta\phi = \frac{2\pi\Delta L}{\lambda} = \frac{2\pi v\Delta t \cos(\theta_i)}{\lambda}$$

- The Doppler shift f_{D_i} in frequency of i th path is given as

$$f_{D_i} = \frac{1}{2\pi} \lim_{\Delta t \rightarrow 0} \frac{\Delta\phi}{\Delta t} = \frac{1}{2\pi} \lim_{\Delta t \rightarrow 0} \frac{2\pi v\Delta t \cos(\theta_i)}{\lambda \Delta t} = \frac{v}{\lambda} \cos(\theta_i) = \frac{v}{c} f_c \cos(\theta_i)$$



Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift calculation

(1) if $-\frac{\pi}{2} < \theta_i < \frac{\pi}{2}$ then Rx moving towards the Tx.
in such case the doppler is positive.

(2) if $\theta_i = 0$ then Tx and the incoming waveform are in opposite direction.
the Maximum Doppler frequency shift when $\theta_i = 0$. In such case

$$f_D = \frac{v}{c} f_c$$

The minimum doppler shift when $\theta_i = 180$

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Example 1: Moving Directly Toward the Base Station ($\theta = 0^\circ$)

- Given:
 - $v = 30$ m/s (Car moving at 108 km/h)
 - $f_c = 2.4$ GHz (WiFi frequency)
 - $\theta = 0^\circ$ (Moving directly toward the transmitter)
- Calculation:

$$f_d = \frac{30}{3 \times 10^8} \times 2.4 \times 10^9 \times \cos(0^\circ)$$

$$f_d = 240 \text{ Hz}$$

Result: $f_d = +240$ Hz (Frequency increases)

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Example 2: Moving Away from the Base Station ($\theta = 180^\circ$)

- Given:
 - $v = 30$ m/s
 - $f_c = 2.4$ GHz
 - $\theta = 180^\circ$ (Moving directly away)
- Calculation:

$$f_d = \frac{30}{3 \times 10^8} \times 2.4 \times 10^9 \times \cos(180^\circ)$$

$$f_d = -240 \text{ Hz}$$

Result: $f_d = -240$ Hz (Frequency decreases)

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Example 3: Moving at a 45° Angle to the Base Station

- Given:
 - $v = 50$ m/s (High-speed train)
 - $f_c = 900$ MHz (Cellular frequency)
 - $\theta = 45^\circ$
- Calculation:

$$f_d = \frac{50}{3 \times 10^8} \times 900 \times 10^6 \times \cos(45^\circ)$$

$$f_d = 106 \text{ Hz}$$

Result: $f_d \approx 106$ Hz (Moderate shift)

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Example 4: Moving Perpendicular to the Base Station ($\theta = 90^\circ$)

- Given:
 - $v = 50$ m/s
 - $f_c = 900$ MHz
 - $\theta = 90^\circ$
- Calculation:

$$f_d = \frac{50}{3 \times 10^8} \times 900 \times 10^6 \times \cos(90^\circ)$$

$$f_d = 0 \text{ Hz}$$

Result: $f_d = 0$ Hz (No Doppler effect)

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Wireless Scenario: Doppler Shift in a Cellular Network

Problem Statement:

A mobile user is traveling on a highway at a speed of **72 km/h** and is connected to a **5G base station** operating at a carrier frequency of **3.5 GHz**. The user is moving at an angle of **60°** with respect to the line connecting the base station and the user.

- Find the Doppler shift experienced by the user.
 - Determine whether the received frequency increases or decreases.
-

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Solution:

Step 1: Given Data

- Speed of the mobile user:

$$v = 72 \text{ km/h} = \frac{72 \times 1000}{3600} = 20 \text{ m/s}$$

- Carrier frequency:

$$f_c = 3.5 \text{ GHz} = 3.5 \times 10^9 \text{ Hz}$$

- Angle of movement:

$$\theta = 60^\circ$$

- Speed of light:

$$c = 3 \times 10^8 \text{ m/s}$$

Small-Scale Fading: Causes

2) Doppler Spread

○ Doppler shift examples

Step 2: Apply Doppler Shift Formula

$$f_d = \frac{v}{c} f_c \cos(\theta)$$

Substituting the values:

$$f_d = \frac{20}{3 \times 10^8} \times (3.5 \times 10^9) \times \cos(60^\circ)$$

Since $\cos(60^\circ) = 0.5$, we simplify:

$$f_d = \frac{20 \times 3.5 \times 10^9}{3 \times 10^8} \times 0.5$$

$$f_d = \frac{70 \times 10^9}{3 \times 10^8} \times 0.5$$

$$f_d = \frac{70}{3} \times 0.5 \times 10^1$$

$$f_d = \frac{35}{3} \times 10$$

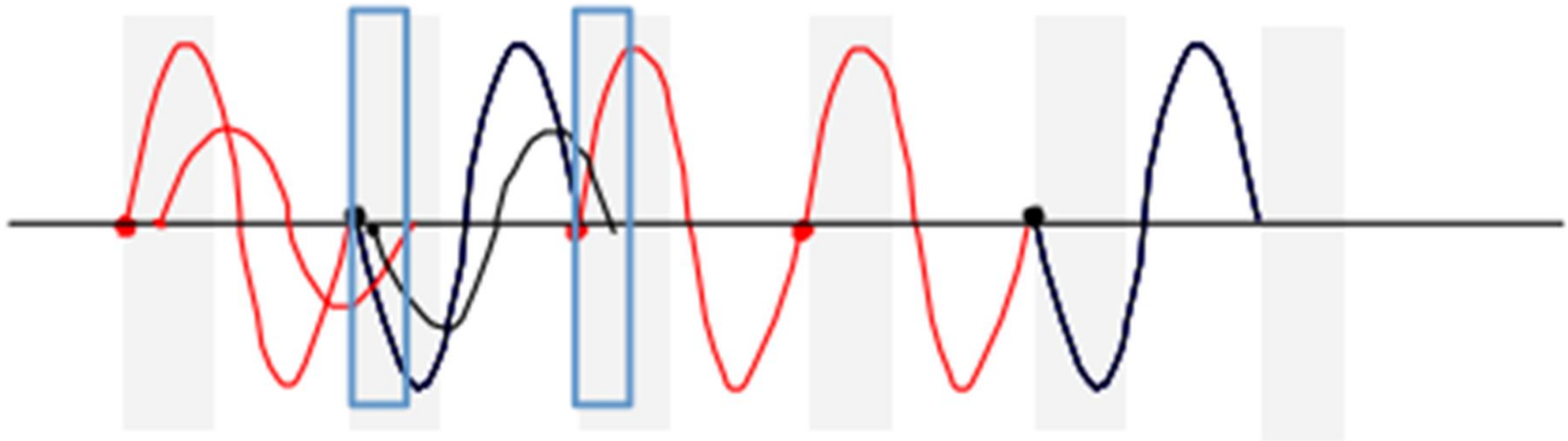
$$f_d \approx 116.7 \text{ Hz}$$

☑ Interpretation

- Since **$\cos(60^\circ)$ is positive**, the Doppler shift is **positive**, meaning the **received frequency increases**.
- This means the user is moving in a direction that has a component toward the base station.

Inter-symbol Interference

- Inter-Symbol Interference (ISI) is a phenomenon in wireless communication where one symbol interferes with subsequent symbols.
- The transmitted signal often reaches the receiver through multiple paths due to reflection, diffraction, and scattering.
- The delayed version of a symbol overlaps with the adjacent symbol.



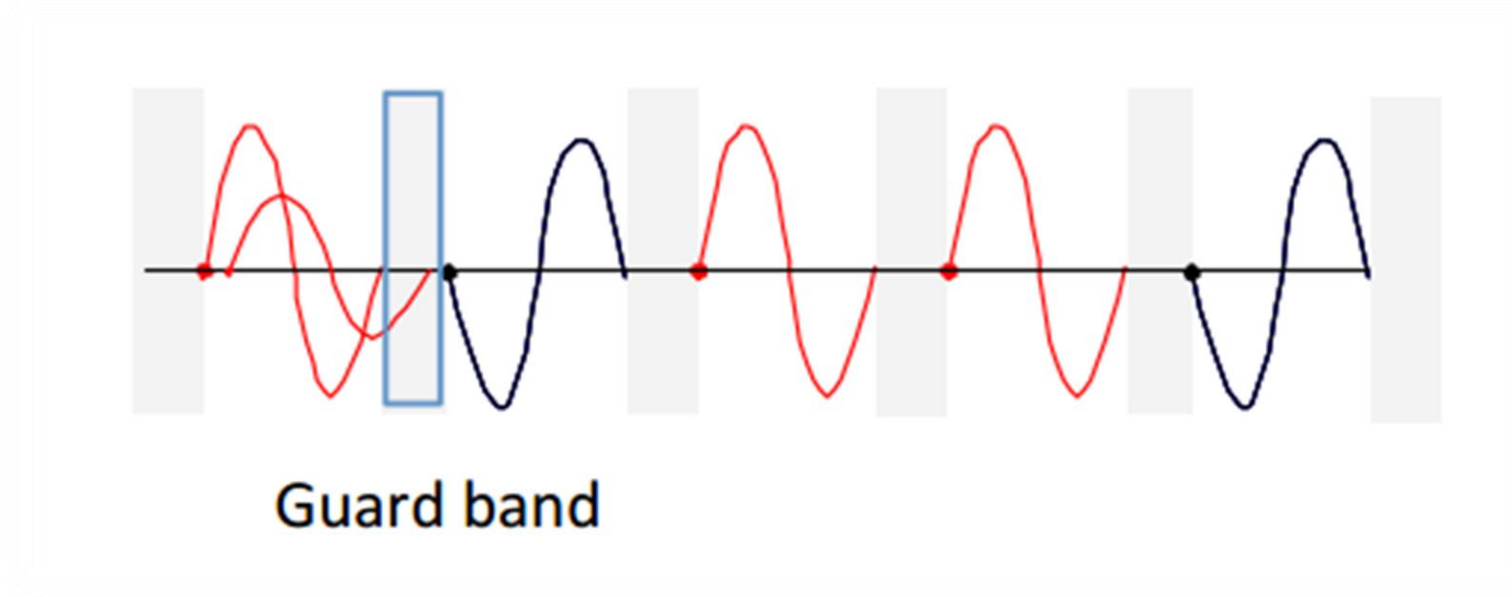
Inter-symbol Interference

■ Impact:

- **Distortion of Signals:** when a symbol overlaps with the adjacent symbols in time, it causes distortion.
- **Reduction in Signal Quality:** ISI decreases the Signal-to-Noise Ratio (SNR) and increases the Bit Error Rate (BER), which reduces the reliability of communication.
- **Difficulty in Detection:** At the receiver, it becomes challenging to determine the correct transmitted symbol, as it is influenced by the echoes or remnants of previous symbols.

Inter-symbol Interference

- **Solution 1:** One simple solution to avoid this is to introduce a guard-band

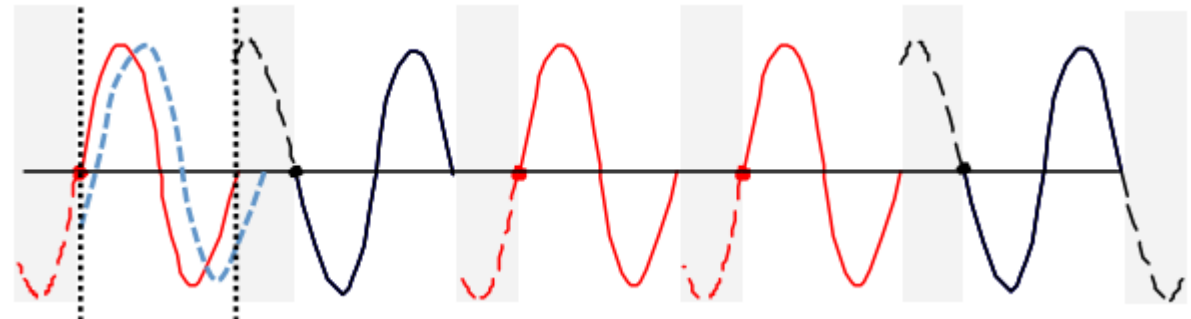
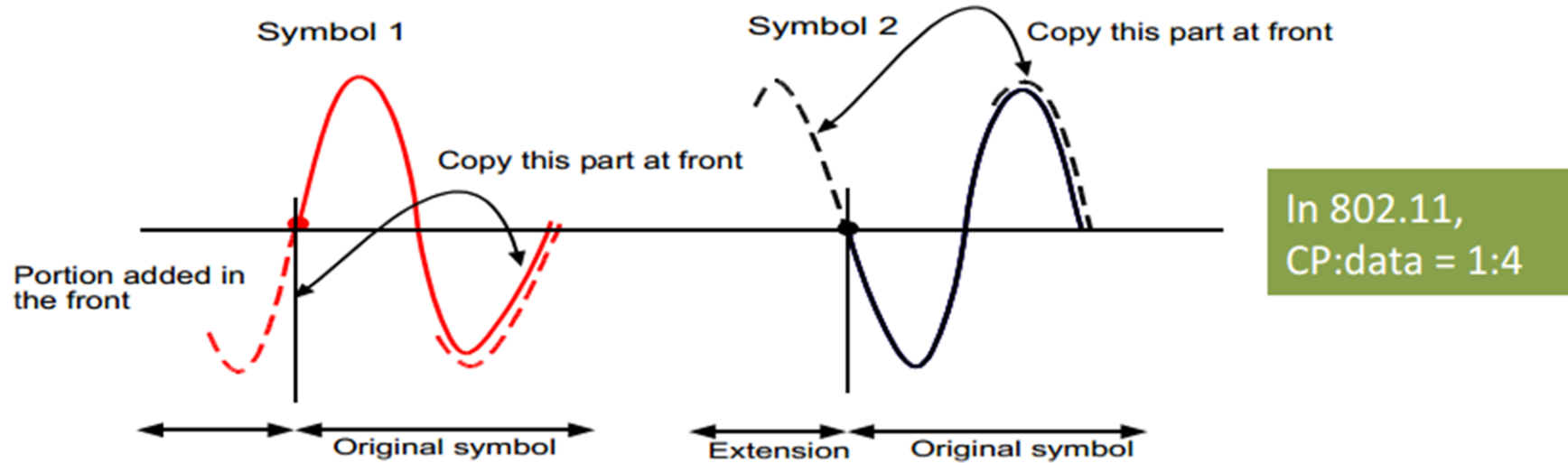


- However, we don't know the **delay spread** exactly.
- The hardware doesn't allow blank space because it needs to send out signals continuously

Inter-symbol Interference

■ Solution 2 : Use Cyclic Prefix

- ▶ Make the symbol period longer by copying the tail and glue it in the front





Thank You